

# Week 1 - Introductory Lecture on Vectors and Matrices

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## 1. Scalars and vectors

A **scalar** is a single number, such as

$$5, \quad -2.4, \quad \pi.$$

A **vector** is an ordered collection of numbers. For example,

$$\mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

is a vector in  $\mathbb{R}^2$ , while

$$\mathbf{w} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$$

belongs to  $\mathbb{R}^3$ .

The number of entries is the vector's **dimension**.

Vectors normally appear as column vectors. A row vector is written as

$$\mathbf{v}^T = \begin{bmatrix} 3 & 2 \end{bmatrix}, \text{ or as } (3, 2)$$

where  $T$  denotes the transpose.

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IMPORTANT POINT: The above "convention", that is of writing vectors as columns, is a (more) modern way of expressing vectors. However, more "traditional" sciences, express vectors as rows. This latter notation we are going to use later, in the problems to solve.

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## Engineering interpretations

A vector can represent many different quantities:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

may describe position in three-dimensional space.

A force vector may be

$$\mathbf{F} = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}.$$

An electrical state vector could be

$$\mathbf{i} = \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix},$$

representing currents in three circuit branches.

A material-composition vector might be

$$\mathbf{c} = \begin{bmatrix} 0.70 \\ 0.20 \\ 0.10 \end{bmatrix},$$

representing fractions of three chemical constituents.

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## 2. Vector addition

Two vectors of the same dimension are added component by component:

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \end{bmatrix}.$$

For example,

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}.$$

Geometrically, vector addition follows the parallelogram or head-to-tail rule

*The following figure about vector addition and subtraction is from Strang's book*

**Represent vector  $v$**    Two numbers   Arrow from  $(0, 0)$    Point in the plane

We add using the numbers. We visualize  $v + w$  using arrows:

*Vector addition* (head to tail)   *At the end of  $v$ , place the start of  $w$ .*

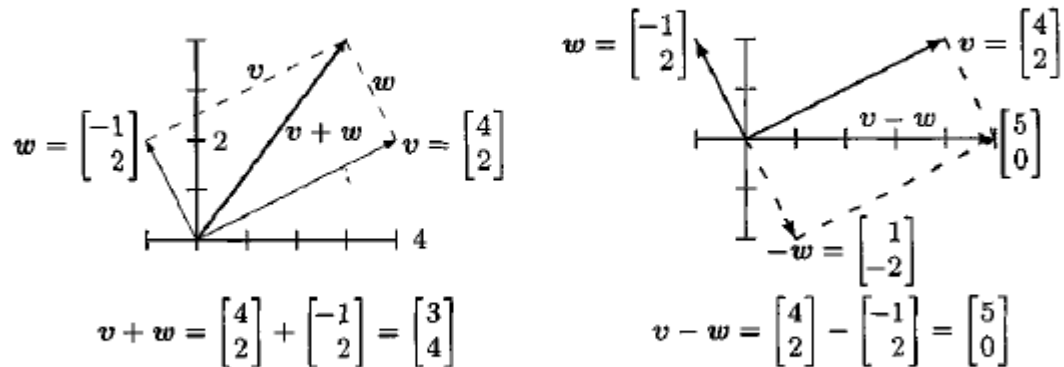


Figure 1.1: Vector addition  $v + w = (3, 4)$  produces the diagonal of a parallelogram. The linear combination on the right is  $v - w = (5, 0)$ .

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## Force example

Suppose two forces act on an object:

$$\mathbf{F}_1 = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \text{ N}, \quad \mathbf{F}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix} \text{ N}.$$

The resultant force is

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 = \begin{bmatrix} 3 \\ 5 \end{bmatrix} \text{ N}.$$

## 3. Scalar multiplication

Multiplying a vector by a scalar multiplies every component:

$$c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}.$$

For example,

$$3 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \end{bmatrix}.$$

Geometrically:

- if  $c > 1$ , the vector becomes longer;
- if  $0 < c < 1$ , it becomes shorter;
- if  $c < 0$ , its direction is reversed;
- if  $c = 0$ , it becomes the zero vector.

The zero vector is

$$\mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

## 4. Linear combinations

A **linear combination** of vectors  $\mathbf{v}_1, \dots, \mathbf{v}_n$  is an expression

$$c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n,$$

where  $c_1, \dots, c_n$  are scalars.

For example, let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

Then

$$2\mathbf{v}_1 - \mathbf{v}_2 = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}.$$

Linear combinations are the foundation of Strang's approach. Many linear-algebra questions have the form:

Can a given vector be produced as a linear combination of some other vectors?

For example, is

$$\mathbf{b} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

a combination of

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}?$$

We seek  $x_1, x_2$  such that

$$x_1 \mathbf{v}_1 + x_2 \mathbf{v}_2 = \mathbf{b}.$$

This gives

$$x_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

Equating components,

$$\begin{aligned} x_1 + 2x_2 &= 5, \\ x_1 - x_2 &= 1. \end{aligned}$$

Solving gives

$$x_2 = \frac{4}{3}, \quad x_1 = \frac{7}{3}.$$

Therefore,

$$\mathbf{b} = \frac{7}{3}\mathbf{v}_1 + \frac{4}{3}\mathbf{v}_2.$$

**A note:** Gilbert Strang's central viewpoint is that linear algebra is fundamentally about **linear combinations**, not merely about manipulating arrays of numbers. Vectors describe quantities with several components, while matrices describe linear transformations and systems of linear equations.

A central expression is

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n,$$

where  $\mathbf{a}_1, \dots, \mathbf{a}_n$  are the columns of  $A$ . This “column picture” connects vectors, matrices and linear systems.

## 5. Length of a vector

For

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

its Euclidean length or norm is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

For example,

$$\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

has length

$$\|\mathbf{v}\| = \sqrt{3^2 + 4^2} = 5.$$

A **unit vector** has length 1. The unit vector in the direction of a nonzero vector  $\mathbf{v}$  is

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

Thus,

$$\hat{\mathbf{v}} = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}.$$

## 6. What is a matrix?

A matrix is a rectangular array of numbers:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

If  $A$  has  $m$  rows and  $n$  columns, then

$$A \in \mathbb{R}^{m \times n}.$$

For example,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

is a  $2 \times 3$  matrix.

Its entry in row  $i$ , column  $j$ , is denoted by  $a_{ij}$ . Thus,

$$a_{23} = 6.$$

## Engineering interpretations

A matrix can represent:

- coefficients of simultaneous equations;
- a geometrical transformation;
- connections in an electrical network;
- stiffness or mass in a mechanical system;
- transition probabilities;
- image pixels;
- experimental observations;
- interactions among variables.

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## 7. Linear independence

Vectors  $\mathbf{v}_1, \dots, \mathbf{v}_n$  are linearly independent if

$$c_1 \mathbf{v}_1 + \cdots + c_n \mathbf{v}_n = \mathbf{0}$$

implies

$$c_1 = \cdots = c_n = 0.$$

For example,

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

are independent.

But

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

are dependent because

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \mathbf{0}.$$

For a matrix  $A$ , its columns are independent precisely when

$$A\mathbf{x} = \mathbf{0}$$

has only the trivial solution

$$\mathbf{x} = \mathbf{0}.$$

## 8. Span and basis

The span of vectors  $\mathbf{v}_1, \dots, \mathbf{v}_n$  is the set of all their linear combinations:

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}.$$

A **basis** is a collection of vectors that:

1. spans the space;
2. is linearly independent.

The standard basis for  $\mathbb{R}^3$  is

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Every vector in  $\mathbb{R}^3$  can be written as

$$x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3.$$

## Exercises

### Exercise 1: Vector operations

Let

$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}.$$

Calculate:

$$\mathbf{u} + \mathbf{v}, \quad 2\mathbf{u} - \mathbf{v}, \quad \mathbf{u} \cdot \mathbf{v}, \quad \|\mathbf{u}\|.$$


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## Exercise 2: Unit vector

Find a unit vector in the direction of

$$\begin{bmatrix} 4 \end{bmatrix}$$


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## Exercise 3: Independence

Determine whether

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

are linearly independent.

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# Short solutions

## Exercise 1

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix},$$

$$2\mathbf{u} - \mathbf{v} = \begin{bmatrix} 3 \\ -6 \\ 8 \end{bmatrix},$$

$$\mathbf{u} \cdot \mathbf{v} = 2 - 4 - 6 = -8,$$

$$\|\mathbf{u}\| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}.$$


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## Exercise 2

$$\|\mathbf{v}\| = 5,$$

so

$$\hat{\mathbf{v}} = \begin{bmatrix} 4/5 \\ -3/5 \end{bmatrix}.$$


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### Exercise 3

The vectors are dependent because

$$\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

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